

```
1  # Python Programming Internship - Week 1
2
3  ## Project Title
4
5  Conditional Statements and Loops in Python
6
7  ## Objective
8
9  The objective of this project is to learn Python control flow using conditional statements and loops.
10
11 ## Tasks Completed
12 |
13 ### 1. Number Classification
14
15 This program checks whether a number is:
16
17 * Even or Odd
18 * Positive, Negative, or Zero
19
20 File: src/number_checks.py
21
22 ### 2. Mathematical Loops and Patterns
23
24 This program:
25
26 * Generates a multiplication table
27 * Calculates the sum of the first N natural numbers
28 * Prints a right-angled triangle pattern using asterisks
29
30 File: src/loop_operations.py
31
32 ## Concepts Used
33
34 * if, elif, else
35 * for loop
36 * User input
37 * Arithmetic operators
38 * Pattern printing
39
40 ## How to Run
41
42 Open the terminal and run:
43
44 python src/number_checks.py
45
46 python src/loop_operations.py
47
48 ## Learning Outcome
49
50 I learned how to use conditional statements and loops in Python to solve basic logical problems and improve
51 problem-solving skills.
```


src new folder

loop_operations.py • README.md .gitignore x

C:\> Users > abhis > Desktop > Python Programming Internship - Week 1 > .gitignore

1 __pycache__/

2 *.pyc

3 .vscode/

4

I

File Edit Selection View Go Run ...

src new folder



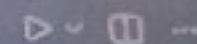
EXPLORER



number_checks.py

loop_operations.py

README.md



CHAT

SESSIONS



SRC NEW FOLDER

loop_operations.py

number_checks.py



> TIMELINE

PROBLEMS

DEBUG CONSOLE

OUTPUT

TERMINAL

PORTS



Enter an integer: 10

The number is Even

The number is Positive

PS C:\Users\abhis\Desktop\src new folder>

number_checks.py > ...

```
1 # Number Classification Program
2
3 num = int(input("Enter an integer: "))
4
5 # Even or Odd
6 if num % 2 == 0:
7     print("The number is Even")
8 else:
9     print("The number is Odd")
10
11
12 # Positive, Negative or Zero
13 if num > 0:
14     print("The number is Positive")
15 elif num < 0:
16     print("The number is Negative")
17 else:
18     print("The number is Zero")
```

Tip: Try the Plan before implementing

+ number_

Describe wh

+ @ A

Local



File Edit Selection View Go Run ...

SRC NEW FOLDER

- loop_operations.py
- number_checks.py

number_checks.py

loop_operations.py > ...

```
2 num = int(input("Enter a number: "))
3
4 print("\nMultiplication Table")
5
6 for i in range(1, 11):
7     print(f"{num} x {i} = {num * i}")
8
9 # Sum of first N natural numbers
10
11 n = int(input("\nEnter N: "))
12
13 total = 0
14
15 for i in range(1, n + 1):
16     total += i
17
18 print("Sum =", total)
19
20 # Star Pattern
21
22 rows = int(input("\nEnter number of rows: "))
23
24 print()
25
26 for i in range(1, rows + 1):
27     print("*" * i)
```

PROBLEMS DEBUG CONSOLE OUTPUT TIMELINE TERMINAL PORTS

Multiplication Table

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```